DGMS Examination 2019 — Question 56 Detailed Solution

Junior Scientific Officer (Geology) | Structural Geology & Rock Mechanics

QUESTION 56

Stress that is generated due to the weight of the overlying rock column in the earth's crust is referred to as:

(a) Confining pressure	(b) Tensile pressure
(c) Lithostatic pressure **CORRECT ANSWER**	(d) Deviatoric pressure

1. Executive Summary & Answer Verification

The correct option is **(c) Lithostatic pressure** (also known as *geostatic pressure* or *overburden pressure*).

In structural geology, geophysics, and petrology:

- The downward gravitational load exerted by a vertical column of rock overlying a point at depth z produces a vertical normal stress:

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\sigma_v = \rho \cdot g \cdot z
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- Over geological time, rocks undergo creep and ductile adjustment, causing horizontal stresses to equalize with vertical stress ($\sigma_h \approx \sigma_v = \sigma_1 = \sigma_2 = \sigma_3$). This equalized, all-sided compressive stress generated specifically by the overlying rock column is formally termed **Lithostatic pressure** (P_{lith}), directly analogous to *hydrostatic pressure* in a body of water.

changes, as at the Moho, would a different value of ρ be required. One need only calculate the pressure to the base of the crust, using an appropriate average crustal density, and continue with depth using a density representative of the mantle.

For example, a reasonable estimate of the average density of the continental crust is 2.8 g/cm^3. To calculate the pressure at the base of 35 km of continental crust, we need only substitute these data into Equation (1), being careful to keep the units uniform (refer to the front inside cover for units and constants):

$$P = \frac{2800\ \text{kg}}{\text{m}^3} \cdot \frac{9.8\ \text{m}}{\text{s}^2} \cdot 35{,}000\ \text{m}$$
$$= 9.6 \times 10^8\ \text{kg/(m s}^2\text{)}$$
$$= 9.6 \times 10^8\ \text{Pa} \approx 1\ \text{GPa}$$

This result is a good average pressure gradient in the continental crust of 1 GPa/35 km, or about 0.03 GPa/km, or 30 MPa/km. Because of (upward) round-off, this gradient is

Figure 56.1: *Lithostatic pressure calculation in the continental crust: For average crustal density $\rho = 2800\text{ kg/m}^3$, pressure at depth $z = 35\text{ km}$ equals $P = \rho g z \approx 1\text{ GPa}$, yielding a lithostatic pressure gradient of ~30 MPa/km. (John D. Winter, Principles of Igneous and Metamorphic Petrology, Chapter 1, Section 7.1, p. 15).*

2. Comprehensive Mechanical Derivations & Stress Decomposition

2.1 Mathematical Formulation of Lithostatic Pressure

If crustal density varies with depth, lithostatic pressure is the integral of density over depth:

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P_{\text{lith}} = \int_0^z \rho(z') \cdot g \cdot dz'
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For a typical continental crust with mean density $\bar{\rho} \approx 2700\text{ to }2800\text{ kg/m}^3$ and $g = 9.81\text{ m/s}^2$:

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\frac{dP_{\text{lith}}}{dz} = \bar{\rho} \cdot g \approx 2750 \times 9.81 =
27,000\text{ Pa/m} = 27\text{ MPa/km} \approx 0.027\text{ GPa/km}
```

At a typical continental Moho depth of 35 km, $P_{\text{lith}} \approx 1.0\text{ GPa}$ (10 kbar).

2.2 Decomposition of Total Stress into Mean and Deviatoric Components

Any general 3D state of stress ($\boldsymbol{\sigma}$) acting on a buried rock element can be decomposed into two distinct mechanical components:

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\sigma_{ij} = \bar{\sigma} \delta_{ij} + \sigma'_{ij} = P_{\text{mean}} \delta_{ij}
+ \sigma_{\text{deviatoric}}
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1. **Mean Normal Stress / Lithostatic Pressure ($P = \bar{\sigma} = \frac{\sigma_1 + \sigma_2 + \sigma_3}{3}$):** An isotropic, all-sided uniform compression. It causes **volume reduction (dilatation)** by closing microcracks and pore spaces, but causes *zero shape change* (no shear deformation).
2. **Deviatoric Stress ($\sigma' = \sigma_{ij} - P \delta_{ij}$):** The non-hydrostatic differential stress component ($\sigma_1 - \sigma_3$). Deviatoric stress produces shear stresses that drive crystal plastic slip, folding, faulting, and rock deformation.

3. Evaluation of All Multiple Choice Options

- **(a) Confining pressure — INCORRECT:** Confining pressure is a general term denoting the all-sided external pressure applied to a rock specimen, commonly used in laboratory triaxial test rigs ($P_c = \sigma_2 = \sigma_3$). While lithostatic pressure acts as a natural confining pressure in the crust, the specific stress generated *by the weight of the overlying rock column* is scientifically designated **Lithostatic pressure**.
- **(b) Tensile pressure — INCORRECT:** Tensile stress involves pulling apart ($\sigma < 0$). Crustal rocks cannot support significant tensile stress at depth.
- **(c) Lithostatic pressure — CORRECT:** Exactly defines the compressive stress generated by the gravitational load of the overlying rock column ($\rho g z$).
- **(d) Deviatoric pressure — INCORRECT:** Deviatoric stress is the directional excess stress responsible for distortion and shape change, not the isotropic overburden weight.

Textbook Authority

John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd ed., Chapter 1, Section 7.1, p. 14–15):
"Pressure increases as a result of the weight of the overlying material... For rock, this equalized pressure is called lithostatic pressure, given by P = ρ g z, with ρ being the density of the overlying rock."
Haakon Fossen, *Structural Geology* (2nd ed., Cambridge University Press):
"The vertical stress due to the weight of the overburden is σv = ρ g z. In a quiet crustal regime where shear stresses have relaxed, the state of stress becomes isotropic (σ1 = σ2 = σ3), known as lithostatic or geostatic pressure."